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Falcon 9 CRS-10 Vehicle Simulator and Navigation

Topics in Aerospace Engineering - Team 5

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1 Introduction and Project Scope

1.1 Project 2 objectives

Project 2 develops a physics-based Falcon 9 simulator and a navigation system driven by simulated sensor measurements. Forces, moments, changing mass properties, and numerical integration generate the vehicle motion; the navigation algorithm estimates states from imperfect observations of that motion. The course identifies this environment as the computational foundation for guidance design in Project 3 and integrated control in Project 4 [11, slides 2–4].

The assignment covers equations of motion and coordinate systems, gravity and atmosphere, propulsion and aerodynamics, mass/inertia variation, configuration changes, and estimation of position, velocity, attitude, and angular velocity. Its validation requirement combines representative flight phases, public flight-data comparisons, and navigation-error analysis [11, slides 4–6]. Accordingly, the report distinguishes physical-model verification, agreement with external flight evidence, and estimation error relative to simulation truth. These comparisons establish different aspects of the simulator's credibility.

1.2 CRS-10 selection and vehicle configuration

CRS-10 is SpaceX's tenth contracted Dragon cargo-resupply mission to the International Space Station. The official mission material identifies a Falcon 9/Dragon launch from Launch Complex 39A, first-stage recovery at Landing Zone 1, and an approximate sequence of cutoff, separation, and restart events [1]. The public technical webcast and its published ascent digitization provide an accessible reference for a quantitative comparison [13, 14].

The mission exercises three connected trajectory segments: common-stack ascent, first-stage return, and upper-stage continuation. It therefore tests configuration changes and inherited states as well as continuous motion. The Dragon payload also fixes the configuration choice: the model carries one loaded Dragon/trunk assembly rather than a generic satellite and fairing. NASA's 2490 kg cargo inventory is a subset of that loaded assembly, not its total mass [2].

Figure 1 identifies the selected hardware and official mission document. The press kit supplies mission context and approximate preflight timing; generic Falcon user guides supply supplementary geometry and operating-point evidence with their configuration/date limitations. Internal dimensions and mass properties require the evidence-qualified modeling described in the vehicle chapter.



Figure 1 CRS-10 reference configuration. Left: Falcon 9 and Dragon at LC-39A before launch, NASA/Glenn Benson [12]. Right: header and mission-patch excerpts from the official SpaceX press kit [1]. The images identify the hardware and source; they do not determine the internal mass distribution.

1.3 Simulation and validation scope

The initial vehicle contains both stages and the loaded Dragon assembly. At separation, a conservative state map creates two children. The first stage proceeds through boostback, coast, entry burn, aerodynamic descent, landing burn, and first geometric contact. The upper child retains Dragon and continues under its own engine and attitude commands. Both children inherit the propagated separation state; a separately selected descent checkpoint does not create the full-return trajectory.

Flight commands provide a repeatable scenario for exercising the physical models. Scheduled reference times are distinguished from state-triggered return conditions, and project-designed return commands are classified as assumptions. The upper continuation demonstrates independent propagation rather than validated orbital insertion. First-stage navigation propagates its own quaternion from noisy gyros in a multiplicative extended Kalman filter (MEKF) with GNSS/LiDAR updates. A simulated star tracker supplies gated attitude observations; acceptance is checked per run. Physical attitude supplies sensor generation, an assumed initial alignment, and the evaluation reference; it is not continuously supplied to the active estimator.

External trajectory comparison uses the retained video-derived ascent observations over approximately 0–140 s. Their classification is E: they are digitized public readouts, not released onboard telemetry. The ascent operating point and pitch timescale were adjusted using the same record, so their agreement is calibration evidence. Model-level tests, continuous state transfer, return/upper trajectories, and navigation errors provide additional evidence with their own stated assumptions. Navigation error against the simulator's truth does not by itself establish the accuracy of the physical vehicle model.

2 Simulation Architecture and Shared Conventions

2.1 Physical truth, measurements, and estimates

The physical branch computes loads and mass properties and integrates vehicle motion. The measurement branch samples that truth through sensor error and availability models. Navigation propagates its own attitude and estimates the IMU-station motion. Comparison uses the same station and time. Figure 2 separates these interfaces.

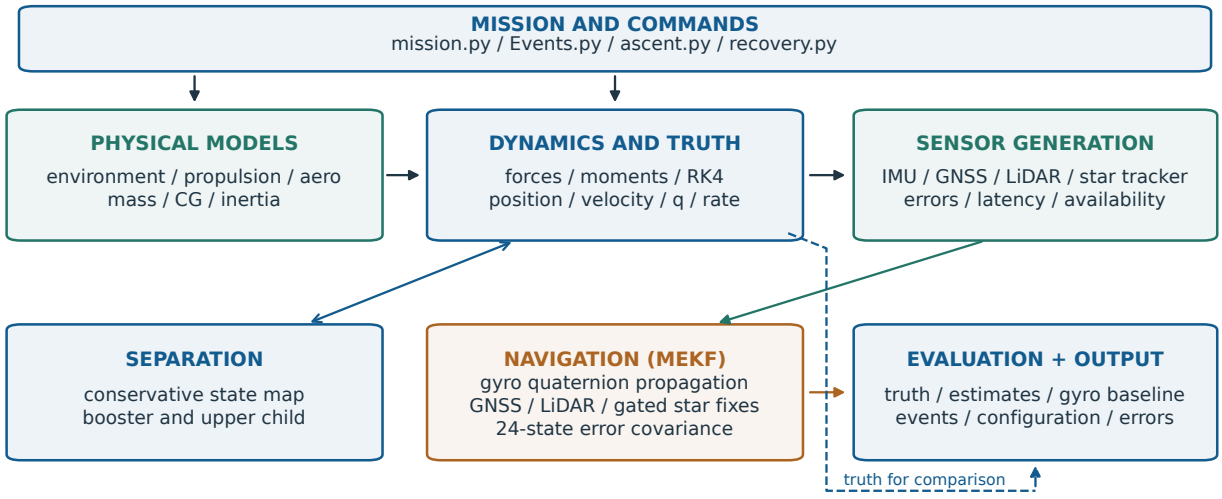


Figure 2 Implemented data flow. The MEKF uses simulated sensor measurements and its own propagated attitude. Truth supplies sensor generation and evaluation; initial alignment is assumed. Return commands use physical-state feedback, while navigation is evaluated separately.

The mission layer selects the initial configuration, commands, and accepted phase. A throttle command becomes thrust and flow through propulsion; an aerodynamic configuration and fin command become force and moment through aerodynamics. The dynamics model then calculates the resulting motion. Neither the force provider nor the mass model prescribes the reference altitude/speed history. Table 1 assigns the corresponding implementation responsibilities.

Function	Main modules	Interface

Vehicle properties	`VehicleData.py`
	`vehicle_mass_model.py`
	`mass_properties.py`
		Trial inventory to mass, CG, inertia, and derivatives.

Environment and loads	`Environement.py`
	`propulsion.py`
	`Aerodynamics.py`
	`load_interface.py`
		Trial state/commands to loads and exhaust data.

Motion and configuration	`dynamics.py`
	`frames.py`
	`staging.py`
		Derivative, integrated state, and inherited child states.

Measurement and estimation	`sensors.py` `kalman.py`	Sensor-station truth to observations and separate estimates.

Mission execution	`main.py`
	`mission.py`
	`Events.py`
		Commands, accepted clocks/events, state synchronization, and saved output.

Table 1 Responsibilities of the shared implementation. Configuration and scenario decisions are kept separate from reusable component arithmetic and equations of motion.

2.2 Frames, origins, and attitude convention

Frame B is right-handed and fixed to the vehicle, with $+x_B$ toward the nose. Frame E is Earth-centred, launch-aligned, and rotating with Earth. Code identifiers call it ECEF, but its axes are not conventional Greenwich-axis ECEF. At the launch site, $+x_E$ is north, $-y_E$ east, and $+z_E$ up; the initial structural base is $[0, 0, R_E]^\mathsf{T}$ with $R_E = 6371000\,\mathrm{m}$.

The scalar-first Hamilton quaternion $\boldsymbol{q}_{BE} = [q_w, q_x, q_y, q_z]^\mathsf{T}$ maps Body components into E using $\mathbf{C}_{BE}$. Equation (1) defines the vector transformation and launch orientation:

$$\boldsymbol{u}^E = \mathbf{C}_{BE}\boldsymbol{u}^B, \qquad \boldsymbol{q}_{BE}(0) = \begin{bmatrix} \sqrt{1/2} & 0 & -\sqrt{1/2} & 0 \end{bmatrix}^\mathsf{T}, \qquad \mathbf{C}_{BE}(0) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}. \tag{1}$$

Thus Body $+x$ is initially upward, Body $+y$ maps to $+y_E$, and Body $+z$ maps to $-x_E$. Quaternions are normalized, and $\boldsymbol{q}$ and $-\boldsymbol{q}$ denote the same orientation. The adopted latitude $\lambda = 28.6°$ rotates the Earth-rate vector into the launch-aligned frame as $\boldsymbol{\Omega}^E = \Omega_E[\cos\lambda, 0, \sin\lambda]^\mathsf{T}$, with $\Omega_E = 7.292115 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$. It does not relocate the pad into Greenwich-axis coordinates.

Reference point and component basis are separate choices. Table 2 distinguishes them. A structural station $\boldsymbol{s}^B$ is measured from the stage base, whereas the corresponding CG force arm is $\boldsymbol{d}^B = \boldsymbol{s}^B - \boldsymbol{c}^B$, where $\boldsymbol{c}^B$ is the current CG station.

Quantity	Meaning
Dynamic position $\boldsymbol{r}_C^B$	Earth centre to instantaneous CG, expressed in current Body axes.
Dynamic velocity $\boldsymbol{v}_C^B$	CG velocity relative to rotating Earth, in current Body components.
Vehicle-object position/velocity	Structural-base state, with the same external origin and Body component basis.
Component/sensor station $\boldsymbol{s}^B$	Fixed structural location measured from the relevant stage base.
Moment and inertia	About the instantaneous vehicle CG, in Body axes.
Filter position/velocity	At the fixed IMU station, expressed in frame E.

Table 2 Shared origins and representations. Storing an Earth-centred vector in Body components does not make its origin the CG.

2.3 Physical and estimated states

The physical state contains 15 stored numbers: three CG-position components, three CG-velocity components, four quaternion components, three angular-rate components, and two stage-indexed propellant inventories. The unit-quaternion constraint leaves three independent attitude coordinates. Each separated child retains the same state layout, with the detached stage's inventory set to zero.

The active MEKF maintains nominal IMU-station position and velocity in E, Body-frame accelerometer bias, the unit quaternion, gyro bias, three gyro scale factors, and six off-diagonal gyro calibration terms. Its covariance represents a 24-component error state: position, velocity, accelerometer bias, small-angle attitude error, gyro bias, scale factors, and misalignment terms. The quaternion occupies four nominal numbers but only three attitude-error coordinates. The original nine-state, supplied-attitude mode remains an optional diagnostic; it is not the default run.

Noisy gyro samples propagate the estimated quaternion using the same Body-to-E convention as Equation (1). GNSS and LiDAR update the coupled estimate; the star-tracker correction is conditional on availability and an innovation gate. Measured rate, estimated gyro bias, and the calibration matrix form the angular-rate estimate; rate is not a separate dynamic filter state. An independent, unaided gyro quaternion provides a drift baseline. The default initial attitude equals the launch alignment, an A-class initialization assumption. The filter does not receive physical attitude during propagation or correction. Calibration-state estimation does not by itself demonstrate identifiability; the nominal run injects gyro bias and noise but zero scale-factor and misalignment errors.

The assumed first-stage stations are $[38, 0, 0]^\mathsf{T}$ m for the IMU, $[41, 0, 0]^\mathsf{T}$ m for GNSS, and $[0.5, 0, 0]^\mathsf{T}$ m for LiDAR. These are A-class experiment settings. They remain on the structure when the CG moves or the upper stage separates. The ideal sensor state follows the structural-point kinematics; sensor errors are added afterward. An estimate at the IMU station is therefore compared with truth at that station, not with the changing geometric CG.

The stored angular rate and simulated ideal gyro use the Body-relative-to-E rate. A physical inertial gyro instead senses Body-relative-to-inertial rotation; the present sensor convention omits that Earth-rate difference. The star-tracker interface explicitly rotates between E and an inertial frame that coincides with E at launch. This conversion does not change the gyro convention. Translation includes Coriolis and centrifugal acceleration; a fully physical inertial strapdown model requires a corresponding gyro Earth-rate treatment.

The star tracker is an A-class simulated aiding sensor, not evidence of CRS-10 flight hardware. Its defaults are 4 Hz, 0.1 s latency, 5 arcsec cross-boresight and 40 arcsec boresight-roll standard deviations, a 60 km minimum altitude, and a 2 deg s^{-1} rate limit. The reviewed seed-1 full-flight run rejects all available star-tracker fixes at the innovation gate. Its measured performance therefore characterizes gyro/GNSS/LiDAR estimation. Sensor and filter chapters must distinguish an implemented correction interface from accepted aiding and assess covariance consistency alongside point-estimate errors.

2.4 Load and execution interfaces

The load record separates nongravitational force in newtons, CG moment in newton-metres, central gravity in $\mathrm{m\,s}^{-2}$, rotating-frame acceleration in $\mathrm{m\,s}^{-2}$, nonnegative stage consumption in $\mathrm{kg\,s}^{-1}$, and nozzle-exit flow data. Station forces contribute $\boldsymbol{d} \times \boldsymbol{F}$ once; free aerodynamic/control moments are supplied separately. Gravity and frame acceleration are added once in the dynamics, rather than included again in the force sum.

At each RK4 trial evaluation, properties and loads are recomputed from the trial state without advancing accepted vehicle inventories, event order, or sensor random generators. The mission loop synchronizes accepted states and samples due sensors. Steps are shortened at relevant command, event, sensor, and output boundaries. At separation, the same first-stage sensor/filter objects continue while two physical children inherit the parent state. The upper child is propagated independently on the common mission clock. Detailed command laws and scheduling belong to mission execution; these interfaces define what the models exchange.

3 Vehicle Configuration, Dynamics, and Mission Definition

3.1 Configuration and evidence-qualified parameters

The attached vehicle comprises stage 1, stage 2, and a loaded Dragon/trunk assembly. After separation, the lower vehicle contains stage 1 and the upper vehicle contains stage 2 and Dragon. The parameter register distinguishes K (authoritative published information), E (literature/data-derived estimate), and A (explicit modeling assumption). E/A denotes a sourced estimate whose use also requires an assumption. Source authority and applicability to the selected 2017 flight are separate considerations.

Table 3 consolidates the configuration used by `vehicle_parameters.py` and `VehicleData.py`. The 62.4 m equivalent height is the sum of adopted stage/Dragon lengths; it is not presented as a measured CRS-10 overall dimension. Stage boundaries and the contiguous attachment datums remain conditional. The 8430 kg loaded Dragon mass is retained as a provisional external estimate; the NASA cargo inventory is included within it. No separate satellite fairing is added.

Parameter	Adopted value	Class	Basis
Outer diameter	3.66 m	K	Later official external geometry [5].
Stage-1 / stage-2 length	42.6/12.6 m	E/A	Legacy FT geometry and assumed station allocation [17].
Stage-1 dry / propellant mass	22200/410900 kg	E	Adopted research mass budget [7].
Stage-2 dry / propellant mass	4000/107500 kg	E	Same literature set [7].
Loaded Dragon assembly	8430 kg	E	Provisional external estimate [3].
Dragon equivalent length / diameter	7.2/3.66 m	E/A	Earlier Dragon geometry transferred to a uniform assembly [4].
Dragon / stage-2 CG station	58.8/48.9 m	A	Midpoints above assumed attached-stage datums.
Stage-1 nominal dry CG	17.521 m	E/A	Component budget and residual allocation.
Initial attached mass	553030 kg	E/A	Sum of the adopted inventories and structures.
Equivalent model height	62.4 m	E/A	Sum of the adopted axial envelopes.

Table 3 Consolidated vehicle inputs. Stations are measured from the attached stack base along Body $+x$. A known external diameter does not establish a known internal tank or mass distribution.

3.2 Propulsion and propellant depletion

Stage 1 has one central engine and eight surrounding engines on an assumed 1.3 m radius ring. Stage 2 has one MVac engine. The force-application pivots are 1.0 m and 0.5 m above the respective bases (A); pivot stations, engine mass-centre stations, and nozzle-exit stations are distinct. Engine dry masses are already included in the stage dry budgets.

For a running steady engine with throttle u, the implementation evaluates Equation (2); an off engine has zero thrust and flow:

$$T = uT_{\mathrm{vac}} - A_e p_{\mathrm{amb}}, \qquad Q = \frac{uT_{\mathrm{vac}}}{I_{\mathrm{sp,vac}} g_0}, \qquad g_0 = 9.80665\,\mathrm{m\,s^{-2}}. \tag{2}$$

Here T is thrust, Q positive propellant outflow, A_e nozzle exit area, p_{amb} ambient pressure, and $I_{\mathrm{sp,vac}}$ the adopted vacuum specific impulse. The code retains a fixed-area ambient-pressure correction; the steady throttle does not scale that pressure term. Smooth ignition/shutdown envelopes subsequently scale the evaluated thrust and flow together. The effective stage-1 vacuum point is $756000 + 0.68(101325) = 824901$ N at full throttle, using the historical sea-level reference and a literature exit area. This mixed-source operating set is E/A, not a certified CRS-10 engine rating [6, 7]. Table 4 lists the adopted engine inputs.

Parameter	Stage-1 engine	Stage-2 engine
Engine count	9	1
Adopted vacuum thrust	824.901 kN (E/A)	935 kN (E)
Vacuum specific impulse	311 s (E)	348 s (E)
Exit area	0.68 m^2 (E)	4.90 m^2 (E)
Steady minimum throttle	0.60 (E)	0.40 (E)

Table 4 Engine operating inputs [7, 6]. The numerical scenario chooses engine count and throttle separately from these engine relations. A stored 5° gimbal parameter (A) does not establish implemented physical gimbal dynamics.

Stage-1 loading is $m_{f,0} = 123500\,\text{kg}$ of RP-1 and $m_{o,0} = 287400\,\text{kg}$ of LOX [9]. Their loaded ratio is 2.32713, while consumption uses $\eta = Q_o/Q_f = 2.35$ [8]. Equation (3) reconstructs both tanks from the stage's total consumed mass Δm:

$$Q_f = \frac{Q}{1+\eta}, \quad Q_o = \frac{\eta Q}{1+\eta}, \qquad m_f = m_{f,0} - \frac{\Delta m}{1+\eta}, \quad m_o = m_{o,0} - \frac{\eta \Delta m}{1+\eta}. \tag{3}$$

The adopted combination depletes LOX first and leaves 1202.128 kg of RP-1 at its mathematical limit. That residual is unusable under the assumed fixed ratio and is separate from a mission reserve. Reconstruction assumes fixed initial loading and no venting, refilling, or variable mixture schedule. Stage 2 retains an equivalent uniformly distributed propellant cylinder rather than the first-stage dual-column model.

3.3 Changing mass, CG, and inertia

3.3.1 Liquid and dry-component models

Each first-stage liquid is a settled, flat-ended cylindrical column at constant density. Table 5 lists its geometry and reference conditions. Structural section lengths are transferred to equivalent tank envelopes; domes, internal hardware displacement, and datum alignment are assumptions.

Quantity	RP-1	LOX
Inner radius [m], E/A	1.8269	1.8271
Bottom station [m], E/A	4.0	18.7
Equivalent section length [m], E/A	14.7	22.1
Density [kg m^{-3}], E	820	1250
Reference temperature [K], E	266	66

Table 5 Equivalent tank inputs. Radii subtract the modeled skin thickness from 1.83 m; section lengths/stations use Ecker's structural model [15, Table 1]. Density/temperature pairs use Miquel's Table 5, whose reference pressure is unspecified [16].

For liquid mass m_i, density ρ_i, radius r_i, bottom station b_i, and outflow Q_i, Equation (4) defines height, centroid, and centroid rate:

$$h_i = \frac{m_i}{\rho_i \pi r_i^2}, \qquad c_i = b_i + \frac{h_i}{2}, \qquad \dot{c}_i = -\frac{Q_i}{2\rho_i \pi r_i^2}. \tag{4}$$

The height must remain inside the adopted envelope. Slosh, ullage motion, reverse-acceleration settling, and feed transients are not resolved. The settled-column approximation is retained through coast and return; restart settling is a separate scenario assumption.

The dry allocation is shell 8000 kg at 21.0 m, fins 1520 kg at 40.3 m, engines 4230 kg at 0.452 m, and folded legs 2000 kg at 6.66 m, using published group estimates with equivalent shapes [8, Table 2]. The remaining 6450 kg closes the 22200 kg budget and is assumed uniformly distributed over 4.0–40.8 m. This yields nominal dry CG 17.521 m. Moving the residual group's centroid anywhere within that interval gives conditional dry-CG bounds 12.18–22.87 m; they are allocation bounds, not statistical confidence limits.

The shell/residual groups use thin-shell equivalents and localized groups use equivalent disks. The liquid columns and loaded Dragon use uniform solid cylinders. Dry-group placements stay fixed; deployed aerodynamic/contact geometry does not separately relocate the dry leg mass. The initial combined vehicle has CG

station 29.3704 m and inertia diagonal approximately $[9.47993 \times 10^5, 1.13857 \times 10^8, 1.13857 \times 10^8]\,\mathrm{kg\,m^2}$. These are model outputs (E/A), not measured CRS-10 mass properties.

3.3.2 Component aggregation and analytical derivatives

A `MassComponent` describes one part or liquid about its own centre. `MassProperties` describes the attached vehicle about its current CG. With component mass m_i, axial centroid c_i, local inertia $\mathbf{J}_{i,c}$, and offset $\boldsymbol{d}_i = (c_i - c)\boldsymbol{e}_x$, Equation (5) gives the combined properties:

$$\begin{aligned} m &= \sum_i m_i, \qquad c = \frac{\sum_i m_i c_i}{m}, \\ \mathbf{J}_C &= \sum_i \left[\mathbf{J}_{i,c} + m_i \mathbf{S}(\boldsymbol{d}_i)\right], \qquad \mathbf{S}(\boldsymbol{d}) = (\boldsymbol{d}^\top \boldsymbol{d})\mathbf{I} - \boldsymbol{d}\boldsymbol{d}^\top. \end{aligned} \tag{5}$$

Here $\mathbf{I}$ is the identity matrix. The implementation differentiates both component properties and parallel-axis offsets. Equation (6) states the first derivatives; second CG/mass derivatives are also evaluated analytically for the motion and sensor interfaces:

$$\begin{aligned} \dot{c} &= \frac{\sum_i (\dot{m}_i c_i + m_i \dot{c}_i) - \dot{m} c}{m}, \\ \dot{\mathbf{J}}_C &= \sum_i \left[\dot{\mathbf{J}}_{i,c} + \dot{m}_i \mathbf{S}(\boldsymbol{d}_i) + m_i \dot{\mathbf{S}}(\boldsymbol{d}_i)\right]. \end{aligned} \tag{6}$$

Axial placement makes the parallel-axis addition $\mathrm{diag}(0, d_i^2, d_i^2)$. Thus Body I_{xx} is axial and I_{yy}, I_{zz} are transverse. Figure 3 shows this same operation in code, and Figure 4 shows its first-stage-only inventory dependence. The late change in CG trend follows the changing weights of the two liquid centroids and the fixed dry allocation.

mass_properties.py | lines 157-169

```
157     inertia = np.zeros((3, 3))
158     inertia_rate = np.zeros((3, 3))
159     for c in components:
160         distance = c.center_axial_m - center
161         distance_rate = c.center_rate_mps - center_rate
162         parallel = np.diag([0.0, distance**2, distance**2])
163         parallel_rate = np.diag([0.0] + [2 * distance * distance_rate] * 2)
164         inertia += c.inertia_at_center_kgm2 + c.mass_kg * parallel
165         inertia_rate += (
166             c.inertia_rate_at_center_kgm2ps
167             + c.mass_rate_kgps * parallel
168             + c.mass_kg * parallel_rate
169         )
```

Figure 3 Exact parallel-axis and derivative accumulation in `mass_properties.py`. Each local inertia is shifted to the combined CG once. Source line numbers and file hash are retained with the figure provenance.

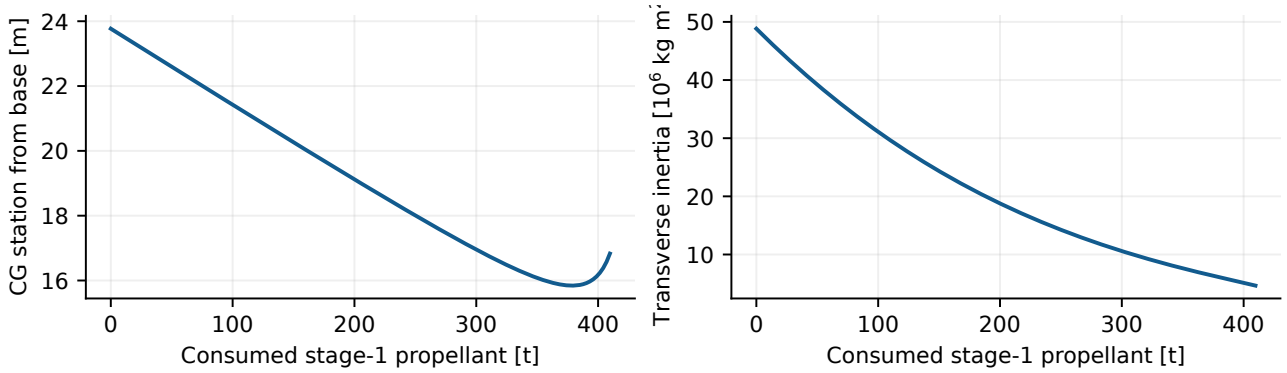


Figure 4 Stage-1-only mass-property sweep from full loading to the fixed-ratio limiting inventory. The curves evaluate the adopted component model against consumed propellant; they are neither measured flight properties nor a separate trajectory simulation.

3.4 Equations of motion and load closure

The changing-mass model uses fixed nozzle-exit stations and balanced transverse outflow. For outlet j, $Q_j \geq 0$ is outward flow, $\mu_j = -Q_j$ its signed mass-removal rate, $\dot{\mu}_j = -\dot{Q}_j$, and $\boldsymbol{d}_j = \boldsymbol{s}_j - \boldsymbol{c}$ the exit arm. $\mathbf{G}_j$ is the outlet's second-moment-per-unit-mass tensor in m^2. The open-system balance retains explicit flow terms, consistent with control-volume rocket dynamics [18].

Equations (7) and (8) are the implemented CG balances. $\boldsymbol{a}_C^B$ is the external-frame coordinate acceleration expressed in Body components; $\boldsymbol{g}^B$ and $\boldsymbol{a}_{\mathrm{rot}}^B$ are supplied separately by the environment:

$$m\boldsymbol{a}_C^B = \boldsymbol{F}^B + m\boldsymbol{g}^B + m\boldsymbol{a}_{\mathrm{rot}}^B + \sum_j \left[2\mu_j(\boldsymbol{\omega}^B \times \boldsymbol{d}_j^B - \dot{\boldsymbol{c}}^B) + \dot{\mu}_j \boldsymbol{d}_j^B\right], \tag{7}$$

$$\mathbf{J}_C\dot{\boldsymbol{\omega}}^B = \boldsymbol{M}_C^B - \boldsymbol{\omega}^B \times (\mathbf{J}_C\boldsymbol{\omega}^B) - \dot{\mathbf{J}}_C\boldsymbol{\omega}^B + \sum_j \mu_j[\mathbf{S}(\boldsymbol{d}_j^B) + \mathbf{G}_j]\boldsymbol{\omega}^B. \tag{8}$$

Pressure-corrected thrust already contains the exhaust reaction; another Qv_e force is not added. The remaining flow terms describe reference migration and angular-momentum outflow. The current closure requires axial CG migration, fixed exits, and balanced transverse exhaust centroids; internal swirl and moving nozzle boundaries are omitted. The code verifies consistency between consumption and mass derivatives before evaluating these balances.

Because the state is stored in current Body axes, Equation (9) includes basis-transport terms. The quaternion uses the same approximate relative-rate convention as Section 2:

$$\begin{aligned} \dot{\boldsymbol{r}}_C^B &= \boldsymbol{v}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{r}_C^B, & \dot{\boldsymbol{v}}_C^B &= \boldsymbol{a}_C^B - \boldsymbol{\omega}^B \times \boldsymbol{v}_C^B, \\ \dot{\boldsymbol{q}}_{BE} &= \tfrac{1}{2}\boldsymbol{q}_{BE} \otimes [0, (\boldsymbol{\omega}^B)^{\mathsf{T}}]^{\mathsf{T}}, & \dot{m}_{p,k} &= -Q_k. \end{aligned} \tag{9}$$

The integrator temporarily uses external translational components within RK4, then returns the accepted state in Body components. This avoids artificial radial interpolation error from rotating an Earth-scale stored vector.

Structural-base and CG states are related by Equation (10), with base O and current CG station $\boldsymbol{c}^B$:

$$\boldsymbol{r}_C^B = \boldsymbol{r}_O^B + \boldsymbol{c}^B, \qquad \boldsymbol{v}_C^B = \boldsymbol{v}_O^B + \boldsymbol{\omega}^B \times \boldsymbol{c}^B + \dot{\boldsymbol{c}}^B. \tag{10}$$

The same transformation applies to a fixed sensor station after replacing the base-to-CG displacement appropriately. `to_state` and `apply_state` use matching flow/rate inputs. Their initial stationary-base condition can therefore have a nonzero geometric-CG velocity from depletion, without implying motion of the sensor mount.

3.4.1 Moment inputs and actuator representation

The moment $\boldsymbol{M}_C^B$ includes aerodynamic moments and an A-class ideal control couple. The active `propulsion.py` relation keeps thrust along Body $+x$ and rejects nonzero gimbal commands; the stored gimbal limit is not an implemented TVC actuator. For `ascent.py`, Equation (11) maps desired Body angular acceleration $\boldsymbol{\alpha}_{\mathrm{cmd}}$ to its added control moment:

$$\boldsymbol{M}_{\mathrm{ideal}} = \mathbf{J}_C\boldsymbol{\alpha}_{\mathrm{cmd}}, \qquad \mathbf{J}_C^{-1}\boldsymbol{M}_{\mathrm{ideal}} = \boldsymbol{\alpha}_{\mathrm{cmd}}. \tag{11}$$

The cancellation in this control contribution is deliberate model inversion, not disappearance of physical inertia or evidence that aerodynamic stability removes the need for TVC. Inertia still determines the requested torque and the response to other loads. The ascent couple is unlimited. `recovery.py` adds gyroscopic feedforward and aerodynamic-moment compensation before fin allocation and moment saturation; changing-inertia and exhaust-flow terms remain in the dynamics.

Thus the scenario demonstrates vehicle propagation with declared moment authority, rather than validated nozzle deflection, gimbal dynamics, or actuator stability margins. A physical TVC model would instead link commanded deflection, actuator limits and dynamics, thrust direction, and the CG moment arm. Flight-command and allocation details belong in the mission-execution chapter; this distinction defines what the moment input represents.

3.5 Initial state and flight-phase definition

The structural base starts at the launch location of Section 2, with zero velocity relative to rotating Earth and zero Body-relative angular rate. The launch quaternion sets Body $+x$ upward; inventories are the full values in Table 3. Commands, configuration transitions, and continuous integration then generate all subsequent states.

Table 6 defines the mission sequence without substituting simulated event times for input conditions. The official schedule labels its entries approximate. Adopting a published time as a simulation command is an A-class choice even when the published schedule itself is authoritative [1].

Transition	Definition in the selected scenario
Liftoff / ascent	Initial attached vehicle; prescribed ascent commands.
Shutdown begins	140 s; project one-second smooth ramp.
MECO / stack coast	141 s; thrust and flow reach zero.
Stage separation	144 s; attached parent, engines off, zero flow and flow derivative.
Upper-stage ignition	152 s; independent upper-child commands.
Boostback ignition	161 s; detached booster and fuel/attitude/readiness guards.
Boostback end / return coast	Miss, fuel, or duration condition followed by smooth shutdown.
Entry ignition	Descending through 45 km with restart guards.
Entry end / aerodynamic descent	$500\,\mathrm{m\,s^{-1}}$ speed target or planning cap, then shutdown.
Landing ignition	Project stopping-clearance prediction with restart guards.
First contact	Lowest assumed deployed leg tip reaches the ground sphere; evaluate landing criteria.
Upper-stage cutoff	545 s; prescribed upper-child shutdown.

Table 6 Physical phase definitions and command references. Entry/landing conditions and return planning criteria are project assumptions. Their actual execution times and numerical results belong to the simulation-results analysis.

First contact uses four assumed leg tips 1.0 m below the base at 3.5 m radius. A successful contact additionally requires deployed legs, vertical speed at most $2\,\mathrm{m\,s^{-1}}$, horizontal speed at most $1\,\mathrm{m\,s^{-1}}$, tilt at most 5°, angular-rate magnitude at most $1\,\mathrm{deg\,s^{-1}}$, and target error at most 10 m. These are A-class project limits, not certified Falcon 9 requirements. The target is an assumed point 14 km south and 6 km east of launch on the ground sphere, rather than surveyed LZ-1 coordinates.

For normalized ramp time $\xi = (t - t_s)/\Delta t$, Equation (12) supplies a smooth shutdown amplitude and its time derivative on $0 \le \xi \le 1$:

$$\gamma = 1 - 10\xi^3 + 15\xi^4 - 6\xi^5, \qquad \dot{\gamma} = -\frac{30\xi^2(1-\xi)^2}{\Delta t}, \qquad \Delta t = 1\,\mathrm{s}. \tag{12}$$

The envelope is constant outside the ramp, with zero endpoint slope and curvature. Scaling thrust and flow together while supplying the analytical flow derivative avoids an unmodeled discontinuous-flow impulse. The duration is a numerical assumption, not a measured engine transient. Return planning budgets, steering laws, and actuator allocation are kept in mission execution rather than embedded in the vehicle equations.

3.6 Separation and inherited child states

The separation map assumes zero impulse and no discarded mass. Both children inherit the parent's attitude and angular rate. For a cut offset $\boldsymbol{\ell}^B$ from the parent structural base, the lower base retains the parent's base state and the upper base follows Equation (13):

$$\boldsymbol{r}^B_{O,u} = \boldsymbol{r}^B_{O,p} + \boldsymbol{\ell}^B, \qquad \boldsymbol{v}^B_{O,u} = \boldsymbol{v}^B_{O,p} + \boldsymbol{\omega}^B \times \boldsymbol{\ell}^B. \tag{13}$$

Upper-child component stations are rebased consistently, while Dragon stays on that child. Different child CG coordinates are expected because the tracked points have changed. Position/velocity continuity is checked at fixed structural points and at the recombined mass centre.

The conservation audit uses angular momentum about the parent CG. With child CG offset $\boldsymbol{\rho}_k$ and velocity difference $\Delta\boldsymbol{v}_k$ from the parent CG, Equation (14) includes spin and displaced-CG motion:

$$\boldsymbol{H}_{\text{after}} = \sum_k \left[\mathbf{J}_k\boldsymbol{\omega}_k + \boldsymbol{\rho}_k \times (m_k\Delta\boldsymbol{v}_k)\right]. \tag{14}$$

The map compares this value with the parent spin angular momentum, and also checks mass, linear momentum, kinetic energy, recombined CG, and structural-point continuity. Relative conservation tolerance is 10^{-10}; position and velocity continuity tolerances are 10^{-8} m and $10^{-8}\,\mathrm{m\,s^{-1}}$; attitude/rate checks use 10^{-12} in their respective norms. These are numerical consistency criteria for the assumed map, not physical separation accuracy.

Afterward, stage 1 excludes stage 2 and Dragon from its mass properties, and the upper child excludes stage 1. Both use inherited states and separate subsequent commands on the same clock. The upper stage retains the uniform propellant/Dragon approximations and its separate drag provider, rather than automatically selecting the declared `second_stack` aerodynamic configuration. Dragon remains attached over the 600 s demonstration interval. The sequence tests continuous configuration/state handling; public validation of the complete return and upper-stage command laws is outside the available ascent reference.

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